

Solutions

6.8 Quiz: Polynomials (no calculator)

1. [Maximum mark: 7]

Let $f(x) = x^2 - 4x - 12$.

(a) Write $f(x)$ in factored form, such that $f(x) = (x - a)(x - b)$. [2]

(b) Write down the x -intercepts of the graph of f . [1]

(c) Write down the y -intercept. [1]

(d) Write $f(x)$ in vertex form, i.e. $f(x) = (x - h)^2 + k$. [1]

(e) Find the coordinates of the vertex of the parabola. [2]

a) $f(x) = (x - 6)(x + 2)$

b) $-2, 6$

c) -12

d) $f(x) = x^2 - 4x + 4 - 16$
 $= (x - 2)^2 - 16$

e) $(2, -16)$

2. [Maximum mark: 7]

Consider the function $g(x) = -(x - 3)^2 + 16$.

(a) Write down the coordinates of the vertex. [1]

(b) Write down the equation of the axis of symmetry. [1]

(c) Show that $g(x) = -x^2 + 6x + 7$. [2]

(d) Solve $g(x) = 0$ to find the x -intercepts. [3]

a) $(3, +16)$

b) $x = 3$

c)
$$\begin{aligned} g(x) &= -(x-3)^2 + 16 \\ &= -(x^2 - 6x + 9) + 16 \\ &= -x^2 + 6x + 7 \end{aligned}$$

d)
$$\begin{aligned} g(x) &= -x^2 + 6x + 7 = 0 \\ &= -(x-7)(x+1) = 0 \\ x &= -1, 7 \end{aligned}$$

SOLUTIONS

3. [Maximum mark: 7]

The line $y = 3x$ and the parabola $y = x^2 - 4$ are drawn on the same axes.

(a) Show that the x -coordinates of the points of intersection satisfy

$$x^2 - 3x - 4 = 0.$$

[2]

(b) Show that $x^2 - 3x - 4 = 0$ has two distinct real roots.

[1]

(c) Factorise $x^2 - 3x - 4$.

[1]

(d) Find the x -coordinates of the points of intersection.

[1]

(e) Find the coordinates of both points of intersection as (x, y) pairs.

[2]

$$a) \quad x^2 - 4 = 3x$$

$$x^2 - 3x - 4 = 0$$

$$b) \quad \Delta = (-3)^2 - 4(1)(-4)$$

$$= 25$$

$$25 > 0 \Rightarrow 2 \text{ real roots}$$

$$c) \quad x^2 - 3x - 4 = (x-4)(x+1)$$

$$d) \quad x = -1, 4$$

$$e) \quad x = -1, y = 3(-1) = -3 \quad (-1, -3)$$

$$x = 4, y = 3(4) = 12 \quad (4, 12)$$

4. [Maximum mark: 7]

The equation $3x^2 - kx + 12 = 0$ has two equal roots.

(a) Write down the discriminant in terms of k . [1]

(b) Find the values of k . [2]

(c) For the positive value of k , find the repeated root. [2]

(d) State the values of k for which the equation has no real roots. [2]

$$\begin{aligned} a) \quad \Delta &= (-k)^2 - 4(3)(12) \\ &= k^2 - 144 \end{aligned}$$

$$\begin{aligned} b) \quad \Delta &= k^2 - 144 = 0 \\ k &= \pm 12 \end{aligned}$$

$$\begin{aligned} c) \quad 3x^2 - (12)x + 12 &= 0 \\ x^2 - 4x + 4 &= 0 \\ (x-2)(x-2) &= 0 \\ x &= 2 \end{aligned}$$

$$\begin{aligned} d) \quad \Delta &= k^2 - 144 < 0 \\ k^2 &< 144 \\ |k| &< 12 \end{aligned}$$

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6.8 Quiz: Polynomials (no calculator)

1. [Maximum mark: 7]

Let $f(x) = x^2 + 6x - 16$.

(a) Write $f(x)$ in factored form, such that $f(x) = (x - a)(x - b)$. [2]

(b) Write down the x -intercepts of the graph of f . [1]

(c) Write down the y -intercept. [1]

(d) Write $f(x)$ in vertex form, i.e. $f(x) = (x - h)^2 + k$. [1]

(e) Find the coordinates of the vertex of the parabola. [2]

a) $f(x) = (x + 8)(x - 2)$

b) $2, -8$

c) -16

d) $f(x) = x^2 + 6x - 16$
 $= x^2 + 6x + 9 - 25$
 $= (x + 3)^2 - 25$

e) $(-3, -25)$

2. [Maximum mark: 7]

Consider the function $g(x) = -(x - 5)^2 + 4$.

(a) Write down the coordinates of the vertex. [1]

(b) Write down the equation of the axis of symmetry. [1]

(c) Show that $g(x) = -x^2 + 10x - 21$. [2]

(d) Solve $g(x) = 0$ to find the x -intercepts. [3]

a) $(5, +4)$

b) $x = 5$

c)
$$\begin{aligned} g(x) &= -(x-5)^2 + 4 \\ &= -(x^2 - 10x + 25) + 4 \\ &= -x^2 + 10x - 21 \end{aligned}$$

d)
$$\begin{aligned} g(x) &= -x^2 + 10x - 21 = 0 \\ &= -(x^2 - 10x + 21) = 0 \\ &= -(x-3)(x-7) = 0 \\ x &= 3, 7 \end{aligned}$$

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3. [Maximum mark: 7]

The line $y = 4x$ and the parabola $y = x^2 - 5$ are drawn on the same axes.

(a) Show that the x -coordinates of the points of intersection satisfy

$$x^2 - 4x - 5 = 0.$$

[2]

(b) Show that $x^2 - 4x - 5 = 0$ has two distinct real roots.

[1]

(c) Factorise $x^2 - 4x - 5$.

[1]

(d) Find the x -coordinates of the points of intersection.

[1]

(e) Find the coordinates of both points of intersection as (x, y) pairs.

[2]

a) $x^2 - 5 = 4x$

$$x^2 - 4x - 5 = 0$$

b) $\Delta = (-4)^2 - 4(1)(-5)$
 $= 36$

$$36 > 0 \Rightarrow 2 \text{ distinct real roots}$$

c) $x^2 - 4x - 5 = (x-5)(x+1)$

d) $x = -1, 5$

e) $x = -1, y = 4(-1) = -4 \quad (-1, -4)$
 $x = 5, y = 4(5) = 20 \quad (5, 20)$

4. [Maximum mark: 7]

The equation $2x^2 - kx + 32 = 0$ has two equal roots.

(a) Write down the discriminant in terms of k . [1]

(b) Find the values of k . [2]

(c) For the positive value of k , find the repeated root. [2]

(d) State the values of k for which the equation has no real roots. [2]

$$\begin{aligned} a) \quad \Delta &= (-k)^2 - 4(2)(32) \\ &= k^2 - 256 \end{aligned}$$

b)

$$k^2 - 256 = 0$$

$$k = \pm 16$$

$$c) \quad 2x^2 - 16x + 32 = 0$$

$$x^2 - 8x + 16 = 0$$

$$(x-4)(x-4) = 0$$

$$x = 4$$

$$d) \quad \Delta = k^2 - 256 < 0$$

$$k^2 < 256$$

$$|k| < 16$$

Sol & James

6.8 Quiz: Polynomials Challenge (no calculator)

1. [Maximum mark: 7]

Consider the function $f(x) = -2(x+2)(x-4)$, for $x \in \mathbb{R}$.

(a) For the graph of f

(i) write down the x -intercepts;

[2]

(ii) find the coordinates of the vertex.

[3]

The function f can be written in the form $f(x) = -2(x-h)^2 + k$.

(b) Write down the value of h and the value of k .

[2]

a) $-2, 4$

ii) $x = \frac{-2+4}{2} = 1$

$$f(1) = -2(1+2)(1-4) \\ = 18 \\ (1, 18)$$

b) $h = 1$
 $k = 18$

2. [Maximum mark: 7]

The line $y = mx$ passes through the origin and is tangent to the parabola $y = x^2 + 9$.

(a) Show that at any point of intersection, $x^2 - mx + 9 = 0$. [1]

(b) Find the discriminant of $x^2 - mx + 9 = 0$ in terms of m . [1]

(c) Find the values of m for which the line is tangent to the parabola. [2]

(d) For each value of m , find the coordinates of the point of tangency. [3]

$$\begin{aligned} \text{a)} \quad x^2 + 9 &= mx \\ x^2 - mx + 9 &= 0 \end{aligned}$$

$$\begin{aligned} \text{b)} \quad \Delta &= (-m)^2 - 4(1)(9) \\ &= m^2 - 36 \end{aligned}$$

$$\begin{aligned} \text{c)} \quad m^2 - 36 &= 0 \\ m &= \pm 6 \end{aligned}$$

$$\begin{aligned} \text{d)} \quad m = 6 \quad x^2 - (6)x + 9 &= 0 \\ (x - 3)^2 &= 0 \\ x = 3, \quad y &= (6)(3) \\ &= 18 \end{aligned}$$

$(3, 18)$

$$\begin{aligned} m = -6 \quad x^2 - (-6)x + 9 &= 0 \\ (x + 3)^2 &= 0 \\ x = -3 \quad y &= (-6)(-3) \\ &= 18 \\ (-3, 18) \end{aligned}$$

Solutions

3. [Maximum mark: 7]

Consider the function $f(x) = 2x^2 - 8x + 3$.

(a) Write $f(x)$ in the form $a(x - h)^2 + k$. [3]

(b) Write down the coordinates of the vertex. [1]

(c) Find the discriminant of $f(x) = 0$. [1]

(d) State the number of x -intercepts of the graph and justify your answer. [2]

(a) $f(x) = 2x^2 - 8x + 3$
 $= 2(x^2 - 4x + 4) - 5$
 $= 2(x - 2)^2 - 5$

(b) $(2, -5)$

(c) $\Delta = (-8)^2 - 4(2)(3)$
 $= 40$

d) Two. $\Delta = 40 > 0$

4. [Maximum mark: 8]

Let $p(x) = 2x^3 + x^2 - 13x + 6$.

(a) Show that $(x + 3)$ is a factor of $p(x)$. [2]

(b) Hence express $p(x)$ in the form $(x + 3)(ax^2 + bx + c)$. [2]

(c) Factorise $p(x)$ completely into three linear factors. [2]

(d) Write down all solutions to $p(x) = 0$. [1]

(e) Write down the y -intercept of the graph $y = p(x)$. [1]

$$\begin{aligned} a) \quad p(-3) &= 2(-3)^3 + (-3)^2 - 13(-3) + 6 \\ &= -54 + 9 + 39 + 6 \\ &= 0 \Rightarrow (x+3) \text{ is a factor} \end{aligned}$$

$$\begin{array}{r} x+3 \overline{) 2x^3 + x^2 - 13x + 6} \\ \underline{2x^3 + 6x^2} \\ -5x^2 - 13x \\ \underline{-5x^2 - 15x} \\ 2x + 6 \end{array}$$

$$p(x) = (x+3)(2x^2 - 5x + 2)$$

$$c) \quad p(x) = (x+3)(2x-1)(x-2)$$

$$d) \quad x = -3, \frac{1}{2}, 2$$

$$e) \quad 6$$

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6.8 Quiz: Polynomials Challenge (no calculator)

1. [Maximum mark: 7]

Consider the function $f(x) = -2(x + 3)(x - 5)$, for $x \in \mathbb{R}$.

(a) For the graph of f

(i) write down the x -intercepts;

[2]

(ii) find the coordinates of the vertex.

[3]

The function f can be written in the form $f(x) = -2(x - h)^2 + k$.

(b) Write down the value of h and the value of k .

[2]

(a i) $-3, 5$

(a ii) $x = \frac{-3+5}{2} = 1$

$f(1) = -2(1+3)(1-5) = 32$

$(1, 32)$

b) $h = 1$

$k = 32$

2. [Maximum mark: 7]

The line $y = mx$ passes through the origin and is tangent to the parabola $y = x^2 + 16$.

(a) Show that at any point of intersection, $x^2 - mx + 16 = 0$. [1]

(b) Find the discriminant of $x^2 - mx + 16 = 0$ in terms of m . [1]

(c) Find the values of m for which the line is tangent to the parabola. [2]

(d) For each value of m , find the coordinates of the point of tangency. [3]

$$\begin{aligned} \text{(a)} \quad x^2 + 16 &= mx \\ x^2 - mx + 16 &= 0 \\ \text{(b)} \quad \Delta &= (-m)^2 - 4(1)(16) \\ &= m^2 - 64 \\ \text{(c)} \quad m^2 - 64 &= 0 \\ m &= \pm 8 \\ \text{d)} \quad m = 8 \quad x^2 - (8)x + 16 &= 0 \\ (x - 4)^2 &= 0 \\ x = 4 \quad y &= (8)(4) = 32 \\ &\quad (4, 32) \end{aligned}$$

$$\begin{aligned} m = -8 \quad x^2 - (-8)x + 16 &= 0 \\ (x + 4)^2 &= 0 \\ x = -4 \quad y &= (-8)(-4) \\ &= 32 \\ &\quad (-4, 32) \end{aligned}$$

Solutions

3. [Maximum mark: 7]

Consider the function $f(x) = 3x^2 + 6x - 2$.

(a) Write $f(x)$ in the form $a(x - h)^2 + k$. [3]

(b) Write down the coordinates of the vertex. [1]

(c) Find the discriminant of $f(x) = 0$. [1]

(d) State the number of x -intercepts of the graph and justify your answer. [2]

(a)
$$\begin{aligned} f(x) &= 3x^2 + 6x - 2 \\ &= 3(x^2 + 2x + 1) - 5 \\ &= 3(x+1)^2 - 5 \end{aligned}$$

(b) $(-1, -5)$

(c)
$$\begin{aligned} \Delta &= 6^2 - 4(3)(-2) \\ &= 60 \end{aligned}$$

d) Two, $\Delta = 60 > 0$

4. [Maximum mark: 8]

Let $p(x) = 2x^3 - 3x^2 - 11x + 6$.

(a) Show that $(x - 3)$ is a factor of $p(x)$. [2]

(b) Hence express $p(x)$ in the form $(x - 3)(ax^2 + bx + c)$. [2]

(c) Factorise $p(x)$ completely into three linear factors. [2]

(d) Write down all solutions to $p(x) = 0$. [1]

(e) Write down the y -intercept of the graph $y = p(x)$. [1]

$$\begin{aligned} (a) \quad p(3) &= \cancel{54} 2(3)^3 - 3(3)^2 - 11(3) + 6 \\ &= 54 - 27 - 33 + 6 \\ &= 0 \Rightarrow x-3 \text{ is a factor} \end{aligned}$$

$$\begin{array}{r} 2x^2 + 3x - 2 \\ x-3 \overline{) 2x^3 - 3x^2 - 11x + 6} \\ \underline{2x^3 - 6x^2} \\ 3x^2 - 11x \\ \underline{3x^2 - 9x} \\ -2x + 6 \\ \underline{-2x + 6} \\ 0 \end{array}$$

$$\begin{aligned} p(x) &= (x-3)(2x^2 + 3x - 2) \\ &= (x-3)(2x-1)(x+2) \end{aligned}$$

$$(d) \quad x = 3, \frac{1}{2}, -2$$

$$(e) \quad 6$$